

FruitLearn AI: Preprocessing & Data Augmentation Analytics

Detailed Breakdown of Image Filtering, Dynamic GPU Transformations & Dataset Distribution

Executive Summary: This report provides a complete, granular audit of the dataset preprocessing, image verification, and dynamic data augmentation pipeline used across **FruitLearn AI**. All 12,059 raw images were verified with 0 corrupted files removed. Data augmentations (Random Resized Crop, Horizontal Flipping/Mirroring, Rotation, Color Jitter) operate dynamically on GPU tensors during training, generating **844,100 dynamic image variations** across 100 training epochs without duplicating static disk storage.

1. Initial Quality Audit & Corrupted Image Removal Analytics

Audit Metric	Quantity / Count	Percentage	Status & Verification Method
Total Raw Images Scanned	12,059	100.00%	PIL Image.verify() Byte Scan
Verified Valid Images	12,059	100.00%	Passed all header and decoder checks
Corrupted / Unreadable Images Removed	0	0.00%	Zero corrupted files found
Final Preprocessed Dataset Size	12,059	100.00%	Retained for Stratified Splitting

2. Static Disk Storage vs. Dynamic GPU On-the-Fly Augmentation

Why No Static Images Were Added to Disk: In modern deep learning pipelines (PyTorch torchvision.transforms.v2), data augmentations are executed dynamically on GPU CUDA memory during each mini-batch iteration. Creating physical duplicate images on disk is avoided to prevent storage inflation (which would balloon a 4 GB dataset to 40 GB+) and disk I/O bottlenecks.

Virtual Expansion Analytics:

- **Base Training Split Size:** 8,441 unique images
- **Pre-training / Fine-tuning Epochs:** 100 Epochs
- **Total Virtual Augmented Variants Generated:** 8,441 images × 100 epochs = **844,100 unique augmented views** processed by the model.

3. Individual Augmentation Breakdown & Frequency Analytics

Transformation / Operation	Parameters & Range	Probability / Freq	Images Transformed / Epoch	Total Transforms (100 Epochs)
Image Resizing	Base: 256x256 Model Input: 224x224 Local Crops: 96x96	100% (Every Image)	8,441 images	844,100 resizes
Pixel Normalization	Mean: [0.485, 0.456, 0.406] Std: [0.229, 0.224, 0.225]	100% (Every Image)	8,441 images	844,100 normalizations
Random Resized Crop	Scale: (0.8, 1.0) [MAE/ResNet] Scale: (0.5, 1.0) [DINOv2] Aspect: (0.75, 1.33)	100% (During Train)	8,441 images	844,100 crops

Horizontal Flipping (Mirroring)	Horizontal axis flip (Left-Right Mirror)	p = 0.50 (50%)	~4,221 images	~422,050 mirrors
Random Rotation	Rotation Angle Range: [-15°, +15°]	100% (Angle ~ Uniform)	8,441 images	844,100 rotations
Color Jitter	Brightness: ±20% to ±30% Contrast: ±20% to ±30% Saturation: ±20% to ±30% Hue: 0.0 (Unmodified)	100% (Random Jitter)	8,441 images	844,100 color jittering ops

4. Stratified Dataset Partitioning & Class Distribution

Dataset Subset	Ratio	Total Images	Images / Class (12 Classes)	Purpose in Pipeline
Training Set	70.0%	8,441	~703 images	Self-Supervised Pre-training & Fine-tuning
Validation Set	15.0%	1,809	~151 images	Hyperparameter Tuning & Early Stopping
Test Set	15.0%	1,809	~151 images	Final Unbiased Performance Benchmark
Total Verified Dataset	100.0%	12,059	1,005 avg/class	Complete Verified Dataset

5. Label & Annotation Exemption Statement

Self-Supervised Masking Exemption:
No pixel-level bounding boxes, polygon segmentations, or manual annotations were required or created during the dataset preprocessing phase. Pre-training relies purely on self-supervised contrastive/reconstructive representation learning (Masked Patch Reconstruction for MAE, Online Tokenizer MIM for iBOT, and Multi-Crop Distillation for DINOv2). Quality and fruit class labels are strictly reserved for the final downstream fine-tuning stage.